PROJECT REPORT

AI-Based Plant Disease Detection Using Image Classification

Comparative Analysis of Custom CNN and MobileNetV2 for Potato Leaf Disease Classification

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Problem Type: Image Classification

Domain: Plant Disease Detection and Agriculture

Models: Custom CNN and MobileNetV2

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1 Introduction

Plant diseases are an important challenge in agricultural production because diseases can reduce crop quality and yield if they are not identified and managed at an early stage. Traditional disease identification generally depends on visual inspection by farmers or agricultural experts, which can be time-consuming and may be affected by human error.

Computer vision and deep learning provide an automated approach for identifying plant diseases from leaf images. Image classification models can learn visual patterns such as spots, discoloration, texture changes, and other disease-related features directly from training images.

This project focuses on automated potato leaf disease classification using deep learning. Three categories are considered:

- Potato Early Blight
- Potato Late Blight
- Potato Healthy

Two different deep learning approaches are investigated: a Custom Convolutional Neural Network (CNN) developed from scratch and MobileNetV2 using transfer learning from ImageNet.

The main purpose of the project is to compare the classification performance of these two approaches on the same potato leaf dataset.

2 Project Objective

The primary objective of this individual project is to develop and evaluate deep learning models for automatic potato leaf disease classification.

The specific objectives are:

1. To prepare a potato leaf image dataset containing three classes.
2. To preprocess and normalize the images for deep learning.
3. To develop a Custom CNN architecture from scratch.
4. To implement MobileNetV2 using ImageNet pretrained weights.
5. To fine-tune the MobileNetV2 model for potato disease classification.
6. To evaluate both models using accuracy, precision, recall, F1-score, confusion matrix, and AUC.
7. To compare the classification performance of the two models.
8. To identify the strengths and limitations of each approach.

3 Dataset Description

The dataset used in this project is the PlantVillage dataset. The original dataset contains 54,305 plant leaf images. From the original dataset, only potato-related images were selected for this project.

The dataset source is:

`https://github.com/spMohanty/PlantVillage-Dataset`

After filtering, the final dataset contained 2,152 potato leaf images belonging to three classes.

Table 1: Potato Leaf Dataset Distribution

Class	Images	Percentage	Label
Potato Early Blight	1000	46.47%	0
Potato Late Blight	1000	46.47%	1
Potato Healthy	152	7.06%	2
Total	**2152**	**100%**	–

The images were converted to RGB format and resized to 224×224 pixels.

3.1 Dataset Characteristics

The dataset is imbalanced because the two disease classes contain 1,000 images each, while the Healthy class contains only 152 images. Therefore, performance on the Healthy class requires particular attention during evaluation.

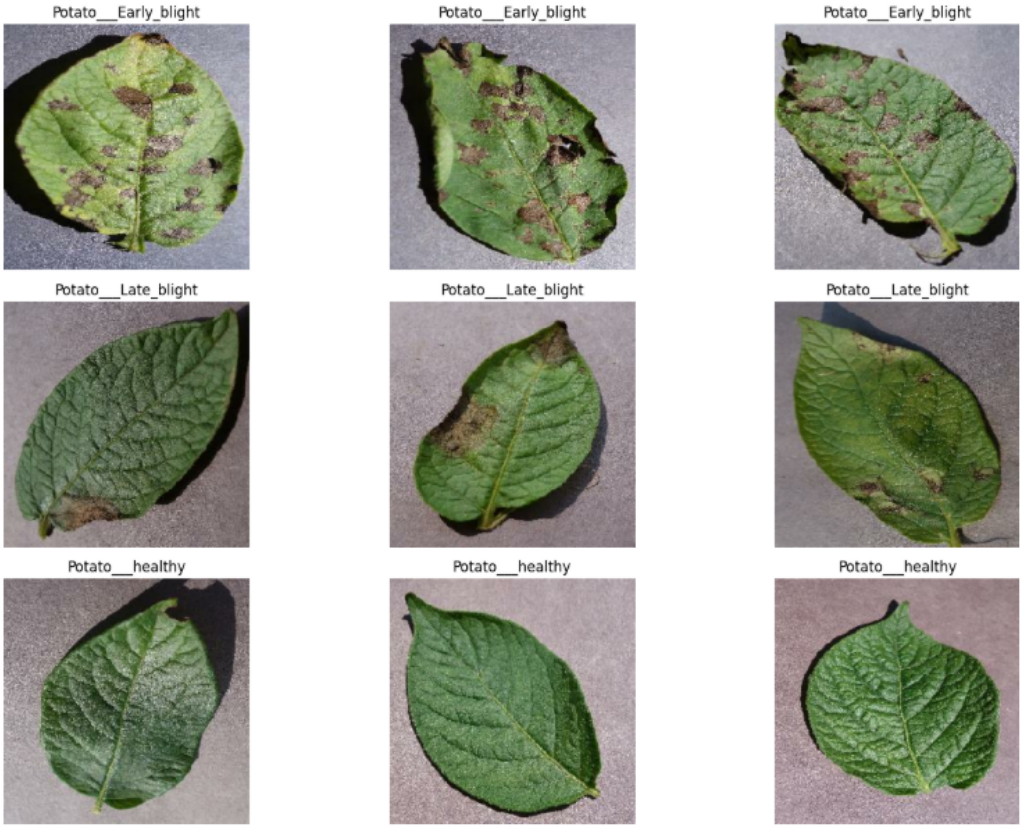


Figure 1: Representative images from the three potato leaf classes.

4 Data Preprocessing

Several preprocessing operations were performed before training the models.

4.1 Image Resizing

All input images were resized to:

$$224 \times 224 \times 3$$

where the three channels represent RGB color information.

4.2 Normalization

Pixel values were normalized from the original range of 0–255 to the range 0–1 using:

$$x_{\text{normalized}} = \frac{x}{255}$$

4.3 Class Filtering

Only the following three potato classes were retained:

- Potato Early Blight
- Potato Late Blight
- Potato Healthy

4.4 Data Splitting

A stratified split with random seed 42 was used to divide the dataset into training, validation, and testing subsets.

Table 2: Dataset Split

Subset	**Images**	**Percentage**
Training	1506	70%
Validation	323	15%
Testing	323	15%
Total	**2152**	**100%**

4.5 Data Augmentation

Random horizontal flipping was applied to the training images as the main augmentation technique. No augmentation was applied to the validation and test sets.

The training data were processed using TensorFlow's `tf.data.Dataset` pipeline with batching, shuffling, and prefetching.

The batch size used for both models was 32.

5 Evaluation Metrics

The models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC.

5.1 Accuracy

Accuracy measures the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

For the multiclass problem, the overall number of correct predictions is divided by the total number of test samples.

5.2 Precision

Precision measures how many samples predicted as a particular class actually belong to that class:

$$Precision = \frac{TP}{TP + FP}$$

5.3 Recall

Recall measures how many actual samples of a class were correctly identified:

$$Recall = \frac{TP}{TP + FN}$$

5.4 F1-Score

The F1-score combines precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

5.5 Macro Average

Macro-averaged metrics calculate the arithmetic mean across the three classes:

$$Macro\ Metric = \frac{Metric_1 + Metric_2 + Metric_3}{3}$$

This is particularly useful for this dataset because the Healthy class contains considerably fewer images than the two disease classes.

5.6 ROC-AUC

The Area Under the Receiver Operating Characteristic Curve (AUC) measures the model's ability to distinguish between classes across different classification thresholds.

AUC values closer to 1.00 indicate stronger class discrimination.

6 Models Used

Two deep learning approaches were implemented and evaluated.

6.1 Custom CNN

The first model was a Custom Convolutional Neural Network developed from scratch.

The architecture consists of three convolutional blocks followed by fully connected layers.

Table 3: Custom CNN Architecture

Layer	**Configuration**
Input	$224 \times 224 \times 3$
Conv2D	32 filters, 3×3, ReLU
MaxPooling	2×2
Conv2D	64 filters, 3×3, ReLU
MaxPooling	2×2
Conv2D	128 filters, 3×3, ReLU
MaxPooling	2×2
Flatten	–
Dense	128 units, ReLU
Dropout	0.5
Output Dense	3 units, Softmax

The Custom CNN contains 11,169,347 total trainable parameters.

The model was trained using:

- Optimizer: Adam
- Learning rate: 0.001
- Loss function: Sparse Categorical Cross-Entropy
- Batch size: 32
- Maximum epochs: 15
- Dropout: 0.5
- Early stopping patience: 10
- Checkpoint monitoring: Validation accuracy
- Random seed: 42

6.2 MobileNetV2

The second approach used MobileNetV2 with transfer learning.

The MobileNetV2 base network was initialized with ImageNet pretrained weights and the original classification head was removed.

The new classification head consisted of:

- Global Average Pooling
- Dense layer with 128 ReLU units
- Dropout with rate 0.5
- Final Dense layer with 3 Softmax outputs

The total model size was 4,374,147 parameters.

6.2.1 Stage 1: Frozen Base Training

During the first stage, the MobileNetV2 base layers were frozen and only the new classification head was trained.

Table 4: MobileNetV2 Head Training Configuration

Parameter	**Value**
Total Parameters	4,374,147
Trainable Parameters	164,355
Non-trainable Parameters	4,209,792
Optimizer	Adam
Learning Rate	0.001
Loss	Sparse Categorical Cross-Entropy
Batch Size	32
Maximum Epochs	10

6.2.2 Stage 2: Fine-Tuning

During the second stage, the last 20 layers of the MobileNetV2 base network were unfrozen and fine-tuned using a smaller learning rate.

Table 5: MobileNetV2 Fine-Tuning Configuration

Parameter	**Value**
Total Parameters	4,374,147
Trainable Parameters	2,367,043
Non-trainable Parameters	2,007,104
Optimizer	Adam
Learning Rate	0.00001
Loss	Sparse Categorical Cross-Entropy
Batch Size	32
Maximum Epochs	5
Unfrozen Layers	Last 20 layers

7 Training Results

7.1 Custom CNN Training

The Custom CNN completed all 15 training epochs. The best recorded epoch was epoch 15.

Table 6: Custom CNN Training Results

Metric	Value
Epochs Completed	15
Best Epoch	15
Training Accuracy	98.01%
Validation Accuracy	97.52%
Training Loss	0.0491
Validation Loss	0.0715

The training accuracy increased throughout training while the loss generally decreased. The validation performance also remained high. Although some validation fluctuations were observed, there was no severe divergence between training and validation performance.

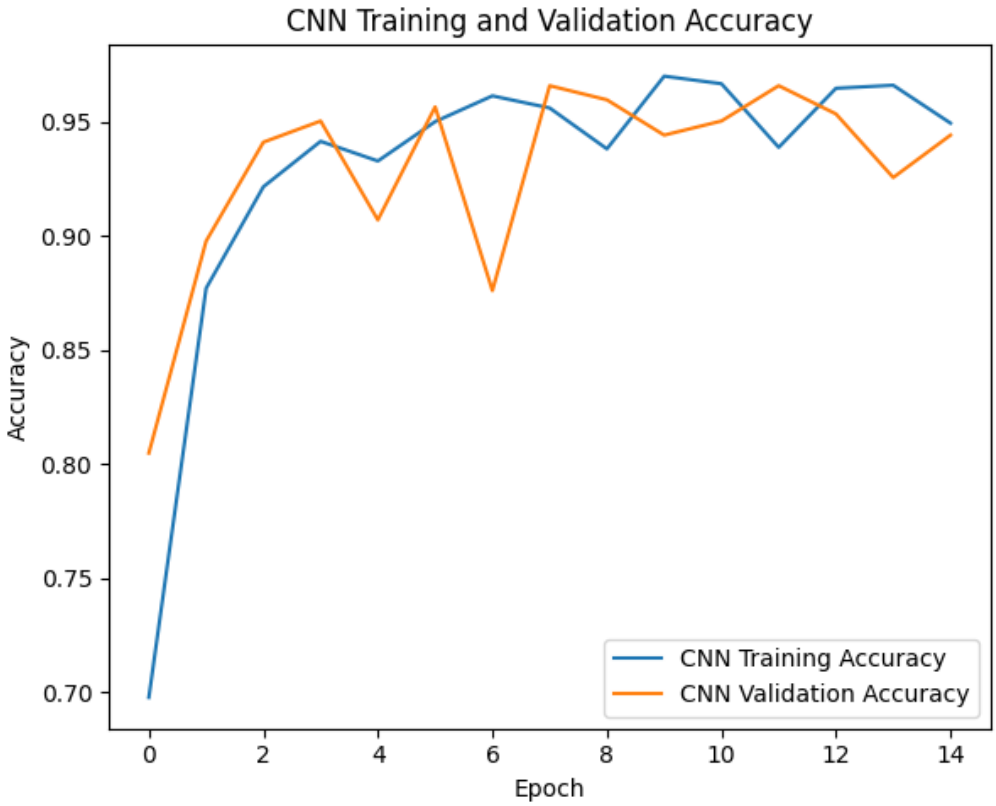


Figure 2: Custom CNN training and validation accuracy.

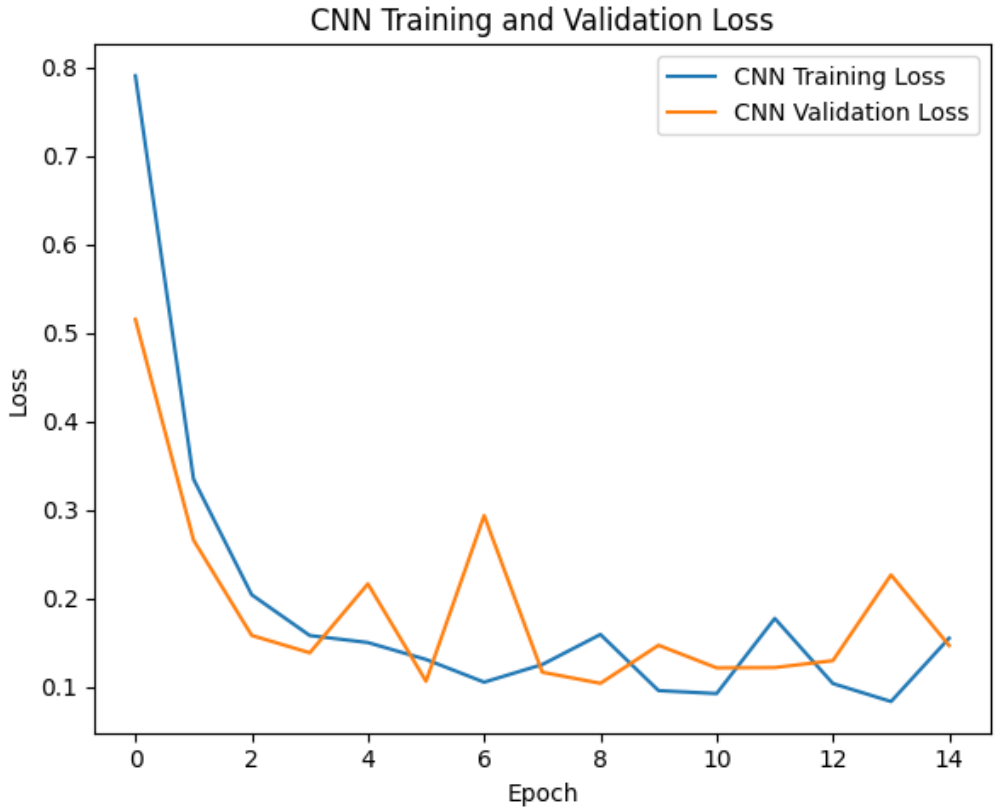


Figure 3: Custom CNN training and validation loss.

7.2 MobileNetV2 Training

During the frozen-base stage, MobileNetV2 completed 10 epochs.

Table 7: MobileNetV2 Head Training Results

Metric	**Value**
Epochs Completed	10
Best Epoch	10
Training Accuracy	98.61%
Validation Accuracy	98.76%
Training Loss	0.0357
Validation Loss	0.0458

The fine-tuning stage was then performed for five epochs.

Table 8: MobileNetV2 Fine-Tuning Results

Metric	**Value**
Epochs Completed	5
Training Accuracy	99.93%
Training Loss	0.0020
Validation Loss	0.0371
Last Recorded Validation Accuracy	99.38%

The fine-tuning stage substantially increased training accuracy. The training loss decreased to 0.0020, while the validation loss remained at 0.0371.

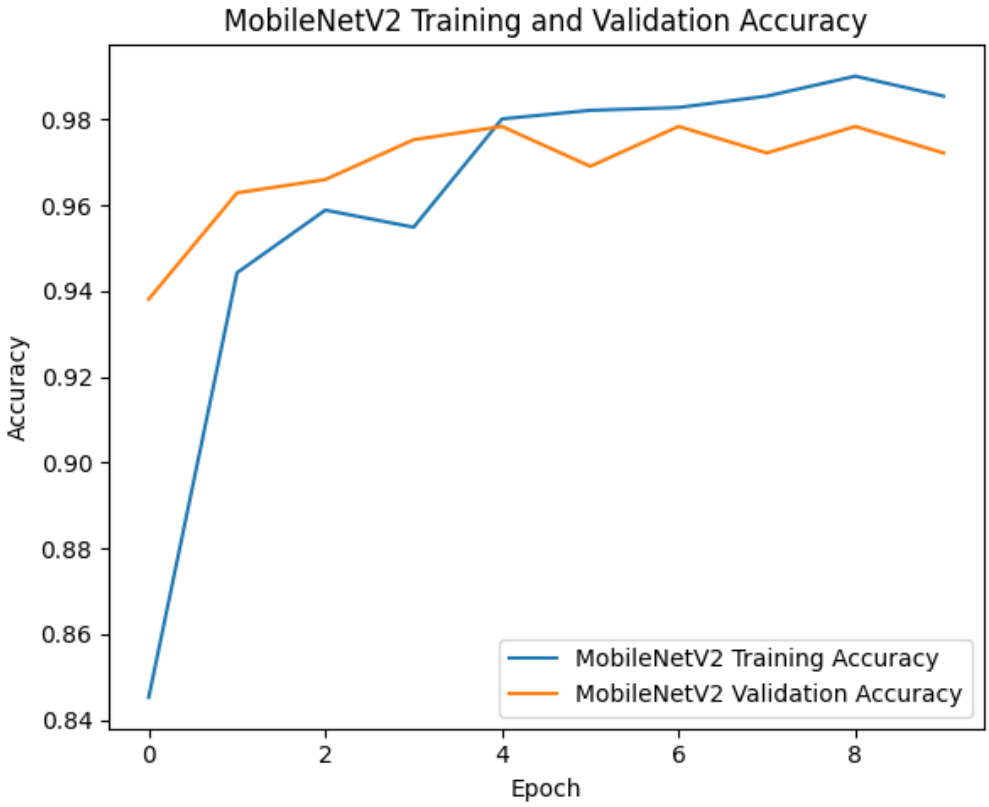


Figure 4: MobileNetV2 training and validation accuracy.

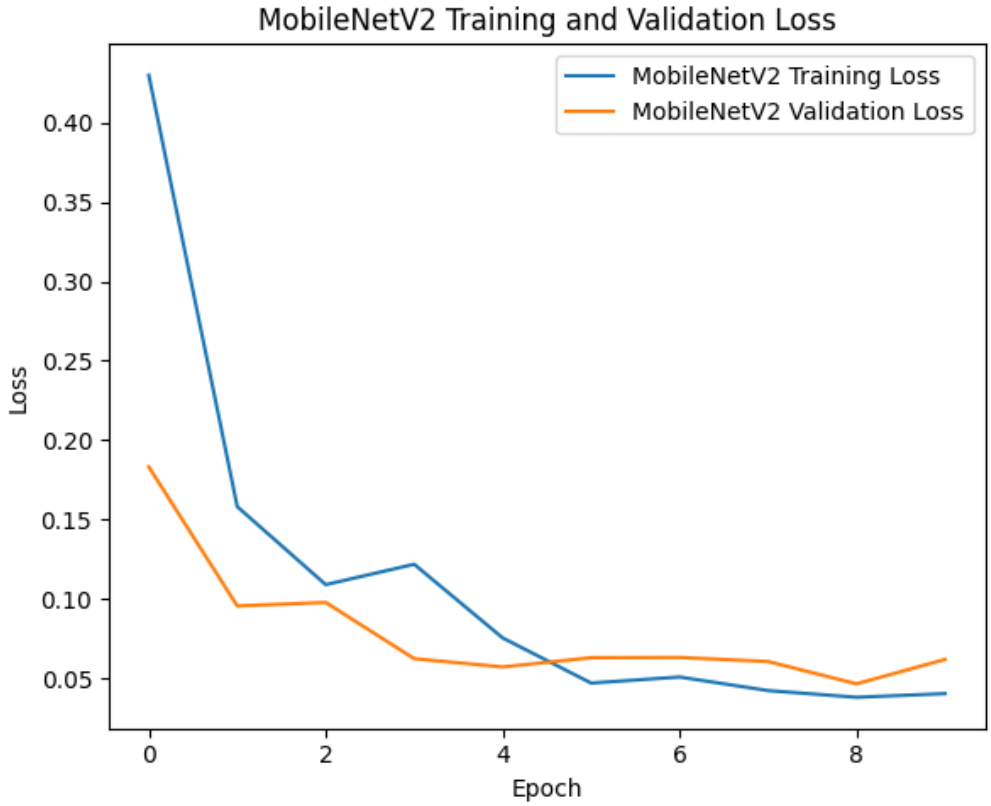


Figure 5: MobileNetV2 training and validation loss.

8 Test Results

The final evaluation was performed on the held-out test set containing 323 images.

Table 9: Overall Test Performance

Metric	**Custom CNN**	**MobileNetV2**
Accuracy	95.67%	97.21%
Macro Precision	94.43%	95.28%
Macro Recall	94.43%	93.09%
Macro F1-Score	94.43%	94.13%

MobileNetV2 achieved a test accuracy of 97.21%, compared with 95.67% for the Custom CNN. MobileNetV2 also achieved higher macro precision.

However, the Custom CNN achieved slightly higher macro recall and macro F1-score. Therefore, the two models show different strengths across the evaluation metrics.

9 Classification Metrics by Class

9.1 Custom CNN

Table 10: Custom CNN Classification Report

Class	Precision	Recall	F1	Support
Early Blight	0.9667	0.9667	0.9667	150
Late Blight	0.9533	0.9533	0.9533	150
Healthy	0.9130	0.9130	0.9130	23
Macro Average	0.9443	0.9443	0.9443	323
Weighted Average	0.9567	0.9567	0.9567	323

9.2 MobileNetV2

Table 11: MobileNetV2 Classification Report

Class	Precision	Recall	F1	Support
Early Blight	0.9867	0.9867	0.9867	150
Late Blight	0.9671	0.9800	0.9735	150
Healthy	0.9048	0.8261	0.8636	23
Macro Average	0.9528	0.9309	0.9413	323
Weighted Average	0.9718	0.9721	0.9718	323

The Healthy class has lower recall and F1-score than the disease classes in both models. This is consistent with the strong class imbalance, since only 23 Healthy images were included in the test set.

10 Worked Metric Calculations

10.1 Custom CNN Accuracy

The Custom CNN confusion matrix contains 309 correct predictions out of 323 test samples.

$$Accuracy = \frac{309}{323} \times 100 = 95.67\%$$

10.2 MobileNetV2 Accuracy

MobileNetV2 correctly classified 314 out of 323 test samples.

$$Accuracy = \frac{314}{323} \times 100 = 97.21\%$$

10.3 MobileNetV2 Healthy Class Recall

For the Healthy class:

$$TP = 19$$

and

$$FN = 4$$

Therefore:

$$Recall = \frac{TP}{TP + FN} = \frac{19}{19 + 4} = \frac{19}{23} = 0.8261$$

Thus, the Healthy class recall is:

$$82.61\%$$

10.4 MobileNetV2 Healthy Class Precision

For the Healthy class:

$$TP = 19$$

and

$$FP = 2$$

Therefore:

$$Precision = \frac{19}{19 + 2} = 0.9048$$

Thus, the Healthy class precision is:

$$90.48\%$$

11 Confusion Matrices

The class order for both confusion matrices is:

$$[\text{Early Blight}, \text{Late Blight}, \text{Healthy}]$$

Rows represent actual classes and columns represent predicted classes.

11.1 Custom CNN Confusion Matrix

The Custom CNN confusion matrix is:

$$\begin{bmatrix} 145 & 5 & 0 \\ 5 & 143 & 2 \\ 0 & 2 & 21 \end{bmatrix}$$

This indicates that:

- 145 Early Blight images were correctly classified.
- 143 Late Blight images were correctly classified.
- 21 Healthy images were correctly classified.
- Five Early Blight images were classified as Late Blight.
- Five Late Blight images were classified as Early Blight.

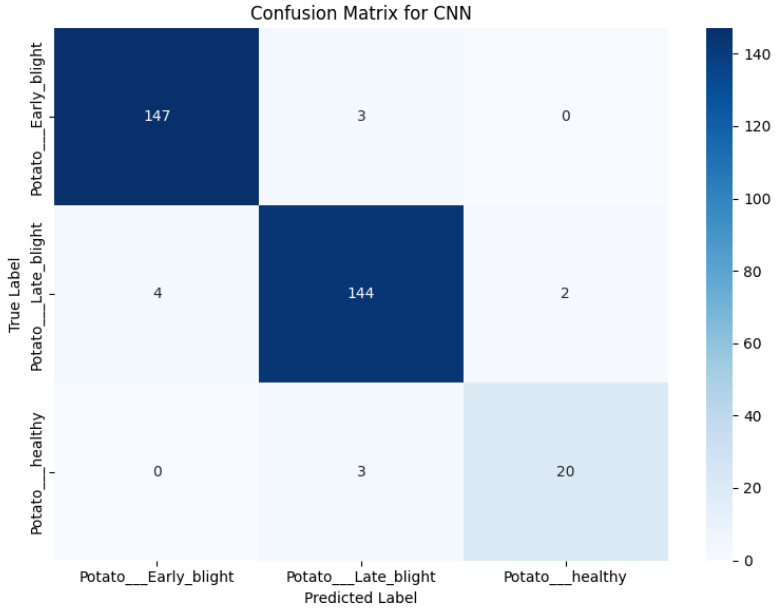


Figure 6: Confusion matrix of the Custom CNN.

11.2 MobileNetV2 Confusion Matrix

The MobileNetV2 confusion matrix is:

$$\begin{bmatrix} 148 & 1 & 1 \\ 2 & 147 & 1 \\ 0 & 4 & 19 \end{bmatrix}$$

The model correctly classified 314 of the 323 test images.

The most notable error is the Healthy class, where four Healthy images were classified as Late Blight.

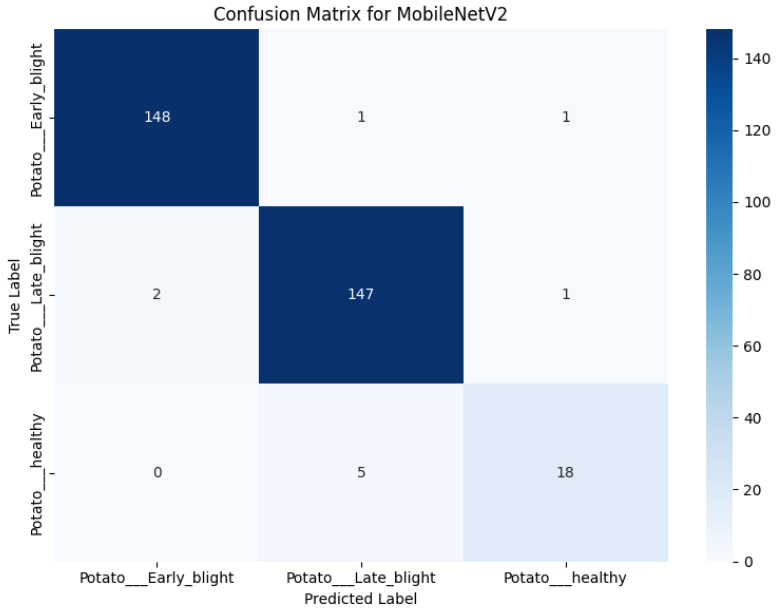


Figure 7: Confusion matrix of MobileNetV2.

12 ROC Curves and AUC Scores

ROC-AUC analysis was performed for each class.

12.1 Custom CNN AUC

The Custom CNN achieved the following AUC values:

Table 12: Custom CNN AUC Scores

Class	AUC
Potato Early Blight	1.00
Potato Late Blight	0.99
Potato Healthy	1.00

The mean AUC is:

$$Mean\ AUC = \frac{1.00 + 0.99 + 1.00}{3} = 0.9967$$

Thus, the Custom CNN achieved a mean AUC of approximately:

$$99.67\%$$

12.2 MobileNetV2 AUC

MobileNetV2 achieved an AUC of 1.00 for all three classes.

Table 13: MobileNetV2 AUC Scores

Class	**AUC**
Potato Early Blight	1.00
Potato Late Blight	1.00
Potato Healthy	1.00

The mean AUC is:

$$Mean\ AUC = \frac{1.00 + 1.00 + 1.00}{3} = 1.00$$

Therefore, MobileNetV2 achieved a mean AUC of 1.00 on the evaluated test predictions.

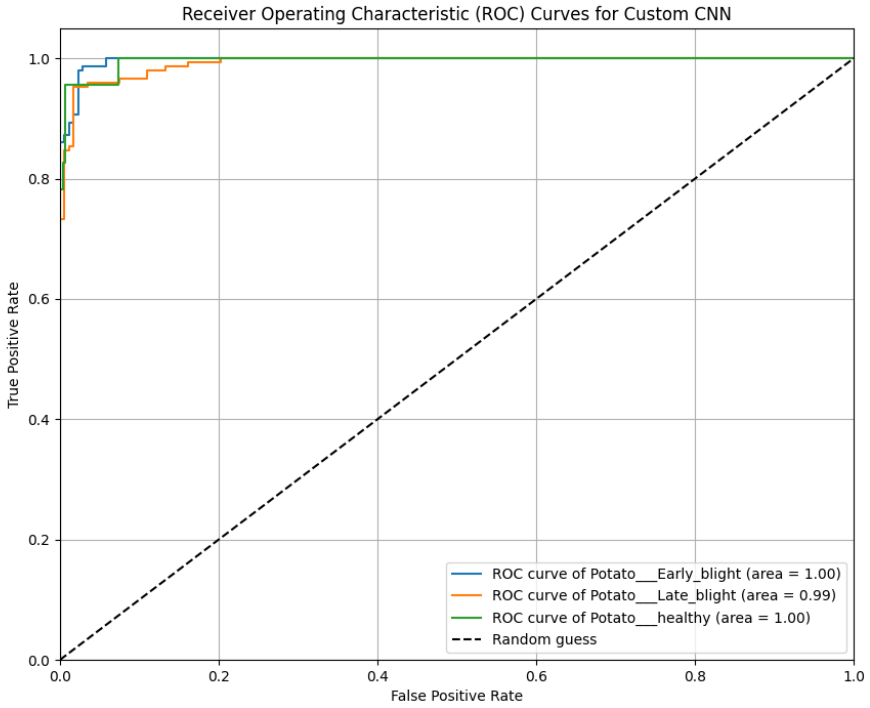


Figure 8: ROC curves and AUC scores for the Custom CNN.

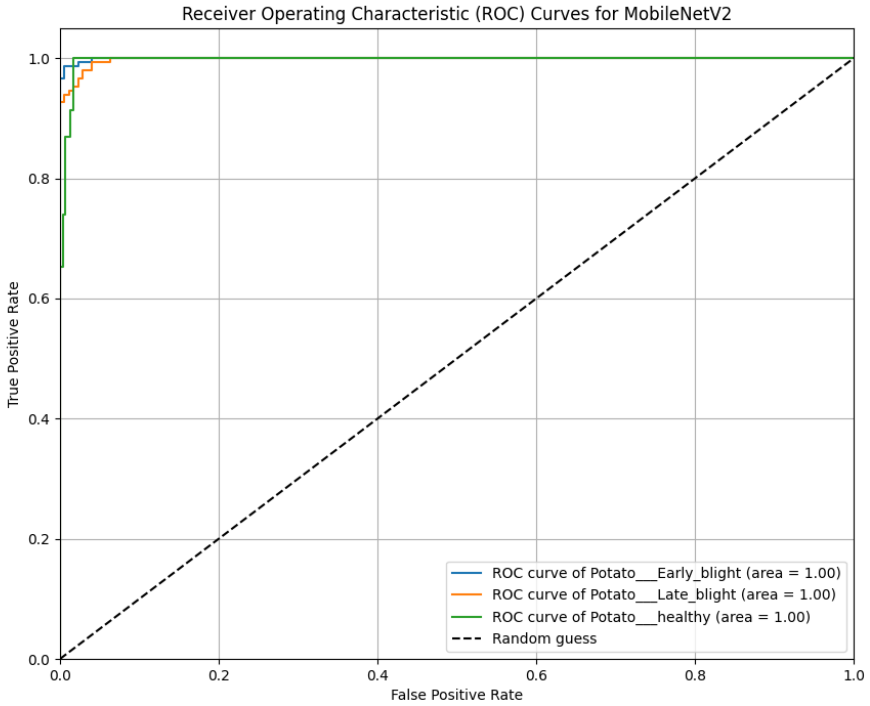


Figure 9: ROC curves and AUC scores for MobileNetV2.

13 Comparative Analysis

The two models were evaluated on the same held-out test set of 323 images.

Table 14: Comparison of Custom CNN and MobileNetV2

Metric	Custom CNN	MobileNetV2
Accuracy	95.67%	97.21%
Macro Precision	94.43%	95.28%
Macro Recall	94.43%	93.09%
Macro F1	94.43%	94.13%
Mean AUC	99.67%	100.00%

13.1 Overall Test Performance

MobileNetV2 achieved the higher overall test accuracy, with 97.21% compared with 95.67% for the Custom CNN.

MobileNetV2 also achieved higher macro precision and mean AUC.

On the other hand, the Custom CNN achieved slightly higher macro recall and macro F1-score.

Therefore, the comparison depends on the evaluation metric. MobileNetV2 demonstrated

stronger overall accuracy and discrimination according to the recorded AUC values, while the Custom CNN maintained slightly higher macro recall and F1-score.

14 Discussion

14.1 Custom CNN

The Custom CNN achieved strong performance despite being trained entirely from scratch.

Its test accuracy was 95.67%, while the macro F1-score was 94.43%. The model also achieved a mean AUC of approximately 0.9967.

The model performed particularly well on Early Blight and Late Blight. Its Healthy-class F1-score was 0.9130, which was higher than the corresponding MobileNetV2 F1-score.

The main advantage of the Custom CNN is that its architecture was designed specifically for the project without relying on pretrained feature representations.

However, the model contains more trainable parameters than the MobileNetV2 configuration used in this project.

14.2 MobileNetV2

MobileNetV2 achieved 97.21% test accuracy and a macro precision of 95.28%.

The model achieved perfect AUC values of 1.00 for all three classes according to the recorded ROC-AUC results.

MobileNetV2 also achieved particularly strong performance for Early Blight and Late Blight.

However, the Healthy class recall was 82.61%, which was lower than the Custom CNN's 91.30%. The model classified four Healthy test samples as Late Blight.

This behavior is important because the Healthy class was substantially underrepresented in the dataset.

15 Training and Validation Observations

The Custom CNN showed a gradual improvement in training accuracy and a corresponding reduction in training loss. Validation performance remained close to training performance, although minor fluctuations were present.

MobileNetV2 achieved high validation accuracy during the frozen-base training stage. During fine-tuning, training accuracy increased substantially and the training loss became very small.

The recorded MobileNetV2 results contain two validation accuracy values from different stages:

- Frozen-base stage: 98.76%
- Last recorded fine-tuning validation accuracy: 99.38%

Because these values correspond to different training stages, they should not be treated as

the same metric or combined without referring to the original training history. The held-out test results are therefore used as the primary basis for the final model comparison.

16 Strengths and Weaknesses

Table 15: Strengths and Weaknesses of the Two Models

Model	Strengths	Weaknesses
Custom CNN	Strong performance from scratch; high macro recall and F1; strong Healthy-class F1	More trainable parameters; performance slightly lower in overall test accuracy
MobileNetV2	Higher test accuracy; higher macro precision; perfect recorded AUC for all classes; transfer learning reduces the need to learn all features from scratch	Lower Healthy-class recall and F1; fine-tuning produces a large gap between training and validation behavior

17 Limitations

Several limitations should be considered when interpreting the results.

1. **Class imbalance:** The Healthy class contains only 152 images compared with 1,000 images for each disease class.
2. **Limited number of classes:** The project considers only three potato leaf categories.
3. **Dataset characteristics:** PlantVillage images are relatively controlled compared with photographs captured in real agricultural environments.
4. **Limited augmentation:** The main training augmentation was random horizontal flipping.
5. **External validation:** The models were evaluated using the held-out PlantVillage test set, but independent field images were not used.
6. **Hyperparameter search:** A systematic hyperparameter optimization process was not performed.
7. **Explainability:** Techniques such as Grad-CAM were not included to visualize the image regions used by the models.
8. **Duplicate verification:** A separate documented duplicate-removal procedure was not included in the recorded preprocessing pipeline.

18 Future Applications and Improvements

Several improvements could be considered for future development:

1. Increase the number of Healthy-class images to reduce class imbalance.
2. Apply class weighting or oversampling techniques.
3. Use stronger and more diverse image augmentation.
4. Evaluate additional transfer-learning architectures such as EfficientNet, ResNet, and DenseNet.
5. Perform systematic hyperparameter optimization.
6. Evaluate the models using independent field-collected potato leaf images.
7. Apply Grad-CAM or similar explainable AI methods to understand model decisions.
8. Investigate model compression, quantization, and pruning for mobile deployment.
9. Expand the system to support additional potato diseases and other crops.
10. Develop a practical mobile or web application for farmers and agricultural workers.

19 Conclusion

This individual project investigated deep learning-based potato leaf disease classification using a Custom CNN and MobileNetV2. A filtered PlantVillage dataset containing 2,152 images from three classes—Early Blight, Late Blight, and Healthy—was used for the experiment.

The Custom CNN achieved a test accuracy of 95.67%, macro precision of 94.43%, macro recall of 94.43%, and macro F1-score of 94.43%. Its recorded AUC values were 1.00 for Early Blight, 0.99 for Late Blight, and 1.00 for Healthy.

MobileNetV2 achieved a higher test accuracy of 97.21% and macro precision of 95.28%. Its macro recall was 93.09% and macro F1-score was 94.13%. The recorded AUC was 1.00 for all three classes.

The results show that MobileNetV2 achieved higher overall test accuracy and macro precision, while the Custom CNN achieved slightly higher macro recall and macro F1-score. The Custom CNN also produced a higher F1-score for the underrepresented Healthy class.

Overall, both approaches demonstrated strong classification performance on the evaluated PlantVillage test set. However, the class imbalance and the controlled nature of the dataset indicate that further evaluation using larger and more diverse field-based datasets would be necessary before applying the models to real-world agricultural environments.